

Three-phase multi-function power meter (DIDO type)

Operating Instruction

I. Product Introduction

The multifunctional power meter (DIDO model) is an upgraded version, featuring enhanced capabilities including switch input, relay output, and transmitter output. This intelligent device supports programmable measurement, display, RS485 digital communication, and electric energy pulse output. It performs comprehensive three-phase electrical parameter measurements (voltage, current, active power, reactive power, apparent power, power factor, frequency) and four-quadrant energy metering, with data display, acquisition, and transmission. Widely applicable in substation automation, distribution automation, smart buildings, and enterprise-level power measurement, management, and evaluation, it achieves 0.5-class measurement accuracy. The system supports LCD on-site display and remote RS-485 digital interface communication via the standard MODBUS-RTU protocol.

The 194F multifunction power meter offers multiple optional expansion features: 4 analog signal outputs, 4 relay outputs (for local electrical parameter alarms or remote control), 4 digital signal inputs, 1 external temperature sensor, and a second RS485 communication port.

II. TECHNICAL PARAMETERS

Parameter		Function
Input MEASUREMENTS	Wiring	Three-phase four-wire / Three-phase three-wire
	Voltage	Range
		AC400V/100V
		Overload
		Sustained: 1.2x, Instantaneous: 2x/1S
		Power dissipation
		<0.3VA
		Impedance
		>1.8MΩ
		accuracy
		RSM measurement, accuracy class 0.2
	Current	Range
		AC5A/1A
		Overload
		Sustained: 1.2x, Instantaneous: 10x/5S
		Power dissipation
		<0.4VA
		Impedance
		<15mΩ
		accuracy
		RSM measurement, accuracy class 0.2
	Frequency	45-65HZ, accuracy ±0.02HZ
	Power	Active power level 0.5, reactive power level 1, and power factor level 0.5
	Electric energy	Active power: 0.5 level, reactive power: 1 level
	temperature	Measurement range: -40 to 120°C, accuracy: ±0.5°C
Output module	Electric pulse output	Passive optocoupler with open-collector output, pulse width 3600
	Communication	RS485 communication interface with MODBUS-RTU protocol, operating at baud rates from 1200 to 19200
	Switch output (optional)	4 The relay outputs with contact capacity of AC250V/3A and DC30V/3A.
	Switch input (optional)	4 input of passive trunk node
	Transmitter output (optional)	4 The transmitter outputs signals in various formats (DC0-20mA, DC4-20mA, DC0-5V, etc.) with a precision of 0.5 class.
Accessory power supply		AC/DC85-265V、AC380V±5%, <5VA
Show		FSTN LCD or LED display
Environment		Operating temperature: -10 to 55°C, Storage temperature: -20 to 85°C
Safe		Insulation: The resistance between signal, power supply, and output terminals to the housing should be >5MΩ. Voltage withstand: The voltage between signal input, power supply, and output terminals should be >2KV.

III. Installation and Wiring

3. 1 Instrument Size

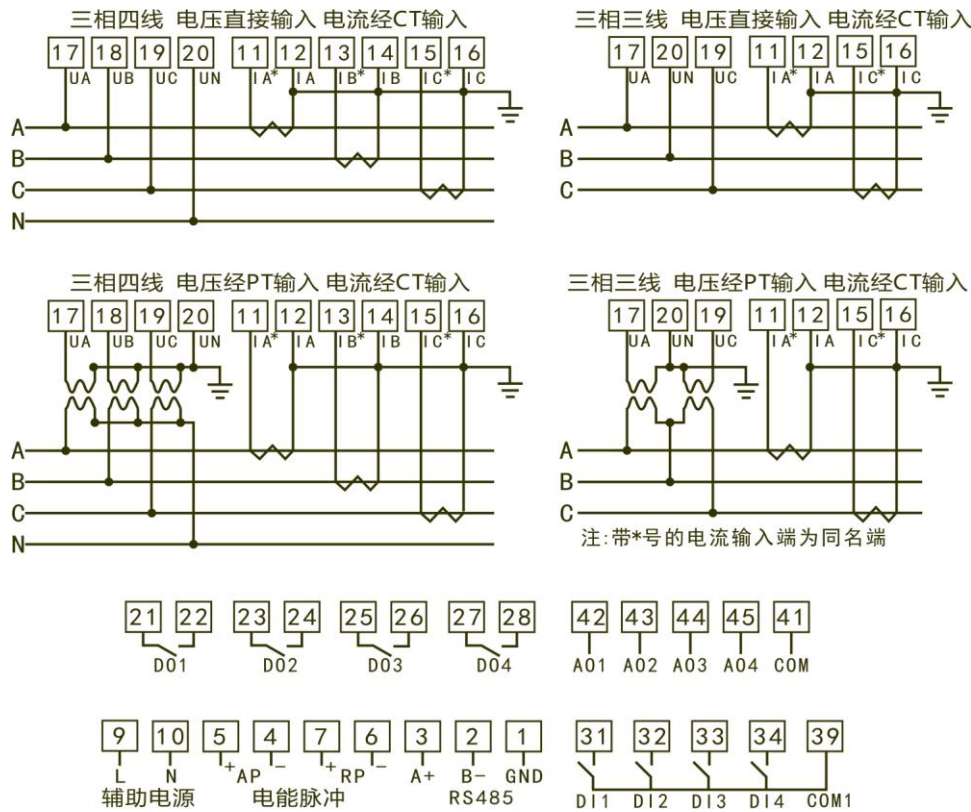
Ideotype Code name	External dimensions (mm)	Opening size (mm)	Minimum installation distance		Degree of depth (mm)
			Horizontal (mm)	Perpendicular (mm)	
42	120×120	111×111	120	120	80
96	96×96	91×91	96	96	80
80	80×80	76×76	80	80	80
72	72×72	67×67	72	72	95

Note: 42.96 supports all optional extended functions; 72 supports optional extended functions: 2-channel switch input and 2-channel switch output; 80 is standard only.

3. 2 Method of Erection

- (1) The size of the hole for opening in the fixed distribution cabinet;
- (2) Remove the instrument, loosen the screws, and detach the mounting bracket.
- (3) The instrument is inserted into the mounting hole from the front;
- (4) Insert the instrument mounting bracket and tighten the screws to secure the instrument.

3. 3 Terminal Wiring



Note: If the wiring diagram differs from the instrument housing diagram, refer to the instrument housing diagram.

(1) Voltage input: The input voltage must not exceed the product's rated input voltage (100V or 400V). If higher, a PT transformer should be used. For easier maintenance, a terminal block is recommended.

(2) Current input terminals: 11, 13, and 15 are the input terminals of the current transformer, where * indicates the current same-name terminal (input terminal). The standard rated input current is 5A; for applications exceeding 5A, an external CT must be used. If the CT is connected to other instruments, the wiring should be connected in series. Before disconnecting the current input wiring, always disconnect the CT's primary circuit or short-circuit the secondary circuit. For easier maintenance, it is recommended to use a terminal block.

(3) Ensure the input voltage and current have consistent phase sequence and direction; otherwise, numerical and sign errors (power and energy) may occur!

(4) The instrument can operate in either three-phase four-wire or three-phase three-wire configurations, with users selecting the appropriate wiring method based on field conditions. Typically, the three-phase three-wire method is used when no neutral line is available, while the three-phase four-wire method is employed when a neutral line exists. The three-phase three-wire configuration requires only two CTs (for phases A and C), whereas the three-phase four-wire configuration necessitates three CTs. The instrument supports two internal wiring modes: the actual wiring configuration and...

The wiring method in the table must be consistent, otherwise the measurement data of the instrument will be incorrect.

(5) 41,42,43,44,45 are analog signal outputs, with 41 being the common terminal of the analog output. 21,22,23,24,25,26,27,28 are switch signal outputs, and 31,32,33,34,39 are switch signal inputs, with 39 being the common terminal of the switch signal input. 4 and 5 are active power pulse outputs, while 6 and 7 are reactive power pulse outputs.

IV. PROGRAMMING OPERATIONS

4. 1 Enter and Exit Programming Mode

Enter programming mode:

To enter the measurement display mode, hold the "SET" button for approximately 3 seconds to access the password authentication screen. Use the ◀, ▲, and ▼ keys to enter the password (the default factory password is 1111). Press the SET button again to enter the programming mode. Important: If pressing the SET button after entering the password exits the measurement display mode, this indicates incorrect password input.

Exit program state:

In the programming state, hold the "SET" key for about 2 seconds to exit the programming state. The user will be prompted to choose whether to save the set value. 8.8.8.8 Save the settings. 8.8.8.8 Do not save the settings.

4. 2 The Use of Key in Programming

Function key SET: Confirm the setting value, enter the next setting, or exit the setting.

Bit selection key ◀: Selects the digital bits to set in a loop. An underline appears below the selected bits.

Add key ▲: Changes the value of the selected digit (the digit value cycles from 0 to 9).

Decrease ▼: Changes the value of the selected digit (the digit value cycles from 9 to 0).

4. 3 Parameter Settings

4. 3. 1 Instrument Operation Parameter Settings (hold the "SET" Key for 3 Seconds to Enter)

Order number	Serial Number Description	Show	Scope
Get into	Enter menu password	8.8.8.8	0000~9999
	Note: Enter the password to access the menu. Only the correct password allows access. The factory preset is 1111		
1	Mode of connection	8.8.8.8	8.8.8.8, 8.8.8.8
	Explain : 8.8.8.8 Three-phase three-wire connection 8.8.8.8 Three-phase four-wire connection (must match the actual wiring method)		
2	Voltage multiplier	8.8.8.8	1~9999
	Note: This parameter sets the PT ratio for the circuit. The factory default is 1. For example, if the circuit uses a 10kV/100V PT, this value should be set to 100.		
3	Current multiplier	8.8.8.8	1~9999
	Note: This parameter sets the CT ratio for the circuit. The factory default is 1. For example, if the circuit uses a 600A/5A CT, this value should be set to 120.		
4	Display mode	8.8.8.8	0~99
	Note: 0 sets the display mode to fixed and manually switches items; 1~99 sets the page to auto-switch, where the value is the auto-switch interval in seconds. Manual switching remains valid in auto-switch mode.		
5	Mailing address	8.8.8.8	1~247
	Note: Instrument address, used to identify the local machine in multi-machine communication		
6	Communication baud rate	8.8.8.8	1200、2400、4800、9600、19200
	Note: Sets the baud rate for RS485 communication. The factory default is 9600.		
7	Communication data format	8.8.8.8	8.8.8.8, 8.8.8.8, 8.8.8.8, 8.8.8.8
	Explain : 8.8.8.8 No parity bit, 8 data bits, 1 stop bit 8.8.8.8 Checksum: 8 data bits and 1 stop bit. 8.8.8.8 Even parity, 8 data bits, 1 stop bit 8.8.8.8 No parity bit, 8 data bits, 2 stop bits		
8	Clear Power 0	8.8.8.8	8.8.8.8, 8.8.8.8
	Explain : 8.8.8.8 Clear power data 0 8.8.8.8 The power data remains unchanged		

9	LCD backlight on time	8.8.8.8	0~9999
	Note: 0 means the LCD backlight is always on; 1 to 9999 indicates the backlight on time in seconds.		
10	Enter password	8.8.8.8	0~9999
	Note: Set a password to access the menu. The default password is 1111.		
11	Relay 1 control item	8.8.8.5	0~255
	Note: 0 indicates remote control. For other settings, refer to the relay operation manual (this menu will not appear if the function is disabled).		
12	Relay 1 lower limit alarm value	8.8.8.8	0~9999
	Note: Refer to the relay operation manual (this menu will not appear if the function is disabled)		
13	Relay 1 upper limit alarm value	8.8.8.8	0~9999
	Note: Refer to the relay operation manual (this menu will not appear if the function is disabled)		
14	Relay 1 alarm return value	8.8.8.8	0~9999
	Note: Refer to the relay operation manual (this menu will not appear if the function is not available)		
15	Relay 1 alarm delay value	8.8.8.8	0~9999
	Note: The unit is seconds. For details, refer to the Relay Operation Manual (this menu is not available if this function is not enabled).		
16	Relay 2 control item	8.8.8.5	0~255
	Note: 0 for remote control. For other settings, refer to the relay operation manual (this menu is not available without this function).		
17	Relay 2 lower limit alarm value	8.8.8.8	0~9999
	Note: Refer to the relay operation manual (this menu item is not available without this function)		
18	Relay 2 upper limit alarm value	8.8.8.8	0~9999
	Note: Refer to the relay operation manual (this menu item is not available without this function)		
19	Relay 2 alarm return value	8.8.8.8	0~9999
	Note: Refer to the relay operation manual (this menu is unavailable if the function is not enabled)		
20	Relay 2 alarm delay value	8.8.8.8	0~9999
	Note: The unit is seconds. For details, refer to the Relay Operation Manual (this menu is not available if this function is not enabled).		
21	Relay 3 control item	8.8.8.5	0~255
	Note: 0 for remote control. For other settings, refer to the relay operation manual (this menu is not available without this function).		
22	Relay 3 lower limit alarm value	8.8.8.8	0~9999
	Note: Refer to the relay operation manual (this menu item is not available without this function)		
23	Relay 3 upper limit alarm value	8.8.8.8	0~9999
	Note: Refer to the relay operation manual (this menu item is not available without this function)		
24	Relay 3 alarm return value	8.8.8.8	0~9999
	Note: Refer to the relay operation manual (this menu is unavailable if the function is not enabled)		
25	Relay 3 alarm delay value	8.8.8.8	0~9999
	Note: The unit is seconds. For details, refer to the Relay Operation Manual (this menu is not available if this function is not enabled).		
26	Relay 4 control item	8.8.8.5	0~255
	Note: 0 for remote control. For other settings, refer to the relay operation manual (this menu is not available without this function).		
27	Relay 4 lower limit alarm value	8.8.8.8	0~9999
	Note: Refer to the relay operation manual (this menu item is not available without this function)		
28	Relay 4 upper limit alarm value	8.8.8.8	0~9999
	Note: Refer to the relay operation manual (this menu item is not available without this function)		
29	Relay 4 alarm return value	8.8.8.8	0~9999
	Note: Refer to the relay operation manual (this menu is unavailable if the function is not enabled)		
30	Relay 4 alarm delay value	8.8.8.8	0~9999
	Note: The unit is seconds. For details, refer to the Relay Operation Manual (this menu is not available if this function is not enabled).		
31	Transmitter Output 1 Control Item	8.8.8.5	0~255

	Note: 0 indicates remote control. For other settings, refer to the transmitter output operation manual (this menu is not available if the function is disabled).		
32	Upper limit value of transmitter output 1	8.8.8.8	0~9999
	Note: Refer to the transmitter output operation manual (this menu item is unavailable without the function).		
33	Transmitter Output 2 Control Item	8.8.8.5	0~255
	Note: 0 indicates remote control. For other settings, refer to the transmitter output operation manual (this menu is not available if the function is disabled).		
34	Upper limit value of transmitter output 2	8.8.8.8	0~9999
	Note: Refer to the transmitter output operation manual (this menu item is unavailable without the function).		
35	Transmitter Output 3 Control Item	8.8.8.5	0~255
	Note: 0 indicates remote control. For other settings, refer to the transmitter output operation manual (this menu item is not available if the function is disabled).		
36	Upper limit value of transmitter output 3	8.8.8.8	0~9999
	Note: Refer to the transmitter output operation manual (this menu item is unavailable without the function).		
37	Transmitter Output 4 Control Item	8.8.8.5	0~255
	Note: Refer to the transmitter output operation manual (this menu item is unavailable without the function).		
38	Upper limit value of transmitter output 4	8.8.8.8	0~9999
	Note: 0 indicates remote control. For other settings, refer to the transmitter output operation manual (this menu is not available if the function is disabled).		
65	Mailing address	8.8.8.8	1~247
	Note: The second RS485 communication line is used for instrument address and local identification in multi-machine communication.		
66	Communication baud rate	8.8.8.8	1200、2400、4800、9600、19200
	Note: The second RS485 communication port is used to set the baud rate. The factory default is 9600.		
67	Communication data format	8.8.8.8	8.8.8.1、8.8.8.4、8.8.8.4、8.8.8.2
	Note: The second communication channel is RS485. 8.8.8.1 No parity bit, 8 data bits, 1 stop bit, 8.8.8.4 Checksum: 8 data bits and 1 stop bit. 8.8.8.4 Even parity, 8 data bits, 1 stop bit. 8.8.8.2 No parity bit, 8 data bits, 2 stop bits		
Withdraw from	Save parameter changes	8.8.8.8	8.8.8.5、8.8.8.8
	Explain : 8.8.8.5 Save parameter changes. 8.8.8.8 The parameter value remains unchanged		

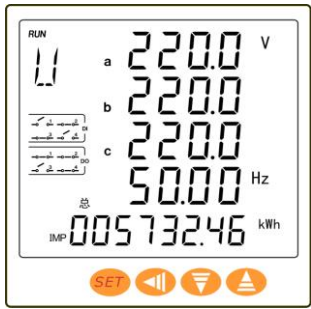
V. PANEL INSTRUCTIONS AND MEASUREMENT INFORMATION DISPLAY

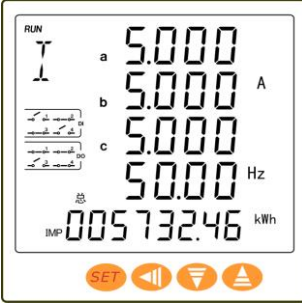
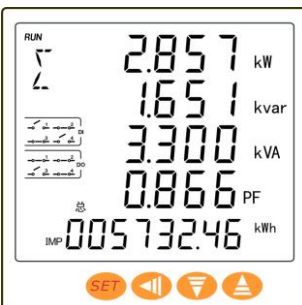
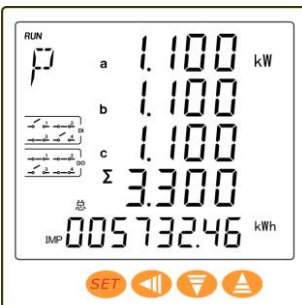
5. 1 Metrical Information

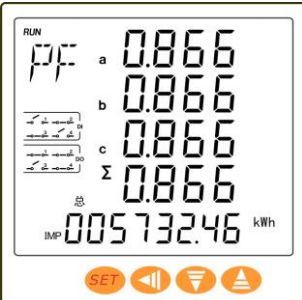
The electrical parameters measured in the power grid include: U_a , U_b , U_c (phase voltage); U_{ab} , U_{bc} , U_{ca} (line voltage); I_a , I_b , I_c (current); P_a , P_b , P_c , P_s (active power per phase and total active power); Q_a , Q_b , Q_c , Q_s (reactive power per phase and total reactive power); P_{Fa} , P_{Fb} , P_{Fc} , P_{Fs} (power factor per phase and total power factor); S_a , S_b , S_c , S_s (apparent power per phase and total apparent power); F (frequency), and active/reactive power. All measured parameters are stored in the instrument's internal data register and can be accessed via the RS 485 communication interface.

5. 2 Display Panel (switch Pages Using the ▲ and ▼ Keys)

5. 2. 1 LCD Display Panel

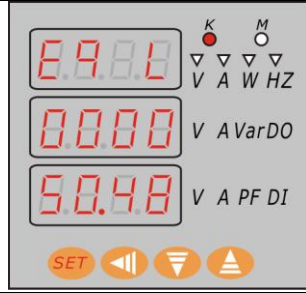
Page	Content	Explain
1 Three-phase voltage and frequency		The first three rows (a, b, c) display phase voltages U_a, U_b, U_c or U_{ab}, U_{bc}, U_{ca}, and the unit is V. The left figure shows $U_a=220.0$ V and $U_b=220.0$ V, $U_c=220.0$ V. In three-phase four-wire input, press the "◀" key to switch between phase and line voltage displays. The fourth row shows frequency $F=50.00$ Hz. The four switches in the DI frame indicate the status of four switch inputs, while the four switches in the DO frame indicate the status of four switch outputs. The bottom row displays the electrical energy information, currently showing the absorption of active power energy at 5732.46 KWh. Switch the display to show the absorption of active power energy by pressing the "SET" key, then release.

		Active power, inductive reactive power, and capacitive reactive power. The power display section operates the same as other display pages. IMP indicates active power absorption, while EX P indicates active power release.  The table method for calculating reactive power energy.  It represents the capacitive reactive power.
2 Three-phase current and frequency		The first three rows (a, b, c) display currents I_a , I_b , and I_c in A, respectively. The left figure shows the $I_a=5.000A$, $I_b=5.000A$, $I_c=5.000A$. The fourth row shows a frequency of 50.00 Hz.
3 There's always some merit to it. Rate, reactive power Power, vision In terms of power, Power factor		The first four rows from top to bottom show the total active power, total reactive power, total apparent power and total power factor. The negative sign in front of the reactive power and power factor indicates capacitive, and the positive sign indicates inductive.
4 Phase active power, total active power		The first three rows (a, b, c) display the active power of Phase A, Phase B, and Phase C respectively, while the fourth row shows the total active power in kW. This interface is not available for three-phase three-wire input.
5 Phase reactive power, total reactive power		The first three rows (a, b, c) display the reactive power of Phase A, Phase B, and Phase C respectively, while the fourth row shows the total reactive power in kVar. This interface is not available for three-phase three-wire input. A negative sign before the reactive power indicates capacitive, while a positive sign indicates inductive.
6 Phase apparent power, total apparent power		The first three rows (a, b, c) display the apparent power of phases A, B, and C respectively, while the fourth row shows the total apparent power in kVA. This interface is not available for three-phase three-wire input.

7 Phase power factor, total power factor		The first three rows (a, b, c) display the power factor of Phase A, Phase B, and Phase C respectively, while the fourth row shows the total power factor. This interface is not available for three-phase three-wire input. A negative sign before the power factor indicates capacitive, while no sign indicates inductive.
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5. 2. 2 LED Display Panel

Page	Content	Explain
1 three-phase voltage		Display the voltage U_a, U_b, U_c (3-phase 4-wire) or U_{ab}, U_{bc}, U_{ca} (3-phase 3-wire), measured in volts (V). When the K indicator light is on, the unit is KV. In the left diagram, $U_a = 220.0V$, $U_b = 220.0V$, $U_c = 220.0V$. For three-phase four-wire input, press the '◀' key to switch between phase voltage and line voltage display.
2 three-phase alternating current		The three-phase currents I_a, I_b, and I_c are displayed in amperes (A). The left figure shows $I_a = 5.000A$ and $I_b = 5.000A$, $I_c = 5.000A$.
3 Active power, reactive power Power, Power Factor		Displays active power P, reactive power Q, and power factor PF. In the left figure, $P = 2.857KW$, $Q = 1.650KVar$, $PF = 0.866$. The highest power factor indicates whether the current reactive power is inductive or capacitive.  Indicates sensibility, with a value of 0.866;  Indicates capacitance, with a value of 0.866.
4 Frequency, switch input/output		The first row displays the frequency $Freq = 50.00Hz$. The second row shows the 1st to 4th switch outputs from right to left, where a connected output is marked as 1 and an open one as 0. The third row displays the 1st to 4th switch inputs from right to left, with a connected input marked as 1 and an open one as 0.
5 active power		The display shows the active power value, with the second row of seven-segment displays showing the high 4 bits and the third row the low 4 bits, forming an 8-bit value. The left figure shows the forward active power value as 300.00KW h. Press the "SET" key to switch between displaying negative active power values. First row . The indicator shows positive active power. . The indicator shows negative active power.

6 reactive power		Display the reactive power value. The second row of the digital tube shows the high 4 bits, and the third row shows the low 4 bits. The left figure shows the inductive reactive power value of 50.48KVarh. Press the "SET" button to switch to display the capacitive reactive power value. The first row .The indicator shows reactive power. .The indicator shows capacitive reactive power.
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VI. FUNCTIONAL MODULES

6. 1 RS485 Communication

6. 1. 1 Physical Layer

- (1) RS485 communication interface, asynchronous half-duplex mode.
- (2) The communication baud rate can be set to 1200,2400,4800,9600, or 19200 bps, with 9600bps as the default factory setting.
- (3) The byte transfer format is as follows: N81 (no parity bit, 8 data bits, 1 stop bit); O81 (odd parity, 8 data bits, 1 stop bit); E81 (even parity, 8 data bits, 1 stop bit); N82 (no parity bit, 8 data bits, 2 stop bits).

6. 1. 2 Modbus-RTU Communication Protocol

This instrument features a serial asynchronous half-duplex RS485 communication interface that operates on the standard MODBUS-RTU protocol, enabling transmission of various data types through the communication line. A single cable can support up to 128 networked instruments, each with customizable communication addresses. The connection must use shielded twisted-pair cables with copper cores and a minimum wire diameter of 0.5mm². During installation, keep communication cables away from high-voltage power lines or strong electric fields. A T-network configuration is recommended, while star or other connection methods are not advised.

The MODBUS protocol employs a master-slave response communication model over a single line. The host computer first addresses a unique terminal device (slave) with a signal, which then transmits a response signal in the opposite direction. MODBUS is designed for communication between host devices (e.g., PCs, PLCs) and terminal devices, not for direct data exchange between standalone terminals. This design prevents terminal devices from occupying communication lines during initialization, as they only respond to query signals received at their own devices.

Host Query: The query message frame contains four components: device address, function code, data information code, and checksum. The address code identifies the slave device to be selected. The function code specifies the required operation—for example, codes 03 or 04 instruct the slave device to read and return register contents. The data segment contains any additional information needed for the function execution. The checksum verifies the frame's integrity, and the slave device implements a CRC16 checksum method to validate the message content.

Slave response: When a slave device generates a valid response, the reply message contains the slave address code, function code, data information code, and CRC16 checksum. The data information code includes the data collected by the slave device, such as register values or status. If an error occurs, the slave device is expected to fail to respond.

The structure of the data frame, i.e., the message format:

Device address	Function code	Data segment	CRC16 check code
1 byte	1 byte	N individual bytes	2 bytes

Device address: Composed of one byte. In our system, only addresses from 1 to 247 are used, with other addresses reserved. Each terminal device's address must be unique, and only the addressed terminal will respond to the corresponding query.

Function code: Specifies the function to be executed by the addressed terminal. The table below lists the function codes supported by this instrument and their respective functions.

Function code	Function
01H	Output status of read relay
02H	Check the status of the digital input
03H/04H	Read the value of one or more registers
05H	Remote control of single relay action
0FH	Remote control of multiple relay actions
10H	Write the value of one or more registers

Data segment:

Contains the data required for the terminal to perform specific functions or the data collected when the terminal responds to queries. The content of this dataUserManual

It can be a number, reference address, or set value.

Checksum: CRC16 occupies two bytes and contains a 16-bit binary value. The CRC value is calculated by the transmitting device and appended to the data frame. The receiving device recalculates the CRC value upon data reception and compares it with the value in the received CRC field. If the two values do not match, an error has occurred.

6. 1. 3 Example of a Communication Message:

(1) Read relay output status (function code 01H): The address of 4 relays is 0~3

Host sends data frame: reads the output status of 4 relays.

Address	Order	Start relay address	Number of relays	Check code
01H	01H	00H,00H	00H,04H	9DH,C9H

Instrument response data frame: Relays 1 and 3 are on, relays 2 and 4 are off

Address	Order	DL	Data segment	Check code
01H	01H	01H	05H	91H,8BH

(2) Check the status of 4 digital inputs (Function code 02H): The addresses range from 0 to 3.

Host sends data frames: reads the status of 4 switch inputs.

Address	Order	Initial switch input address	Number of switch signals	Check code
01H	02H	00H,00H	00H,04H	79H,C9H

Instrument response data frame: Switches 1 and 2 are on, switches 3 and 4 are off.

Address	Order	DL	Data segment	Check code
01H	02H	01H	03H	E1H,89H

(3) Read data register (function code 03H/04H): The host sends a data frame to read three-phase current values.

Address	Order	Start address (high bit first)	Number of registers (high bits first)	Check code
01H	04H	00H,1AH	00H,03H	91H,CCH

Instrument response data frame: Ia=5.000A, Ib=4.996A, Ic=4.980A

Address	Order	DL	Data segment (6 bytes)	Check code
01H	04H	06H	13H,88H,13H,84H,13H,74H	CBH,95H

(4) Remote control of single relay action (function code 05H): The address of 4 relays is 0~3

Host sends data frame: Remote control relay 2 is turned on

Address	Order	Relay address	Relay action value	Check code
01H	05H	00H,01H	FFH,00H	DDH,FAH

Instrument response data frame:

Address	Order	Relay address	Relay action value	Check code
01H	05H	00H,01H	FFH,00H	DDH,FAH

(5) Remote control of multiple relay actions (Function code 0FH): Addresses 0-3 for 4 relays

Host sends data frames: Set relays 1 and 3 to conduct, and relays 2 and 4 to non-conduct.

Address	Order	Start relay address	Number of relays	Data byte count	Relay action value	Check code
01H	0FH	00H,00H	00H,04H	01H	04H	3FH,55H

Instrument response data frame:

Address	Order	Start relay address	Number of relays	Check code
01H	0FH	00H,00H	00H,04H	54H,08H

(6) Write data register (function code 10H):

The host sends data frames: setting the current transformer ratio CT=300 and the voltage transformer ratio PT=100.

Address	Order	Start address	Number of registers	Data byte count	Data segment	Check code
01H	10H	00H,02H	00H,02H	04H	00H,64H,01H,2CH	33H,E4H

Instrument response data frame:

Address	Order	Start address	Number of registers	Check code
01H	10H	00H,02H	00H,02H	E0H,08H

6. 1. 4 Modbus Communication Register Address Table

Attribute: R/W indicates read-write, R indicates read-only.

Address	Item description	Data type	attribute	Explain
0	Enter menu password	Short	R/W	Range: 0 to 9999
1	Electrical parameter display	Short	R/W	High byte. See the menu settings for details.
	Input signal wiring method			Low byte, 0: three-phase three-wire, 1: three-phase four-wire
2	PT voltage transformer	Short	R/W	Range: 1 to 9999
3	Current transformer	Short	R/W	Range: 1 to 9999
4	Mailing address	Short	R/W	High byte, range: 1~247
	Communication baud rate			Low byte, 0: 1200bps~4: 19200bps
5	Communication data format	Short	R/W	0:N81、1:O81、2:E81、3:N82
6	LCD backlight on time	Short	R/W	0~9999
16	Clear power data 0	Short	R/W	Writing any value to this register will clear all power data in 55AAH to 0.
19	Switch input port status	Short	R	See the postscript to Table (1)
	Relay output status		R/W	See the postscript to Table (1)
20	A phase voltage	Short	R	See the postscript to Table (2)
21	B phase voltage	Short	R	See the postscript to Table (2)
22	C phase voltage	Short	R	See the postscript to Table (2)
23	AB phase voltage	Short	R	See the postscript to Table (2)
24	CA phase voltage	Short	R	See the postscript to Table (2)
25	BC phase voltage	Short	R	See the postscript to Table (2)
26	A phase current	Short	R	See the postscript to Table (3)
27	B phase current	Short	R	See the postscript to Table (3)
28	C phase current	Short	R	See the postscript to Table (3)
29	Power, Power Factor Symbol Bit	Short	R	See the postscript to Table (4)
30	Active power of phase A	Short	R	See the postscript to Table (5)
31	Active power of phase B	Short	R	See the postscript to Table (5)
32	C-phase active power	Short	R	See the postscript to Table (5)
33	Active power	Short	R	See the postscript to Table (5)
34	A-phase reactive power	Short	R	See the postscript to Table (5)
35	B phase reactive power	Short	R	See the postscript to Table (5)
36	C phase reactive power	Short	R	See the postscript to Table (5)
37	Total reactive power	Short	R	See the postscript to Table (5)
38	A mutual power	Short	R	See the postscript to Table (5)
39	B-Phase Mutual Power	Short	R	See the postscript to Table (5)
40	C-phase power	Short	R	See the postscript to Table (5)
41	Total apparent power	Short	R	See the postscript to Table (5)
42	A phase power factor	Short	R	See the postscript to Table (6)
43	B phase power factor	Short	R	See the postscript to Table (6)
44	C phase power factor	Short	R	See the postscript to Table (6)
45	Total power factor	Short	R	See the postscript to Table (6)
46	Frequency	Short	R	See the postscript to Table (7)

47~48	Active power (integer part)	Long	R/W	See the postscript to Table (8)
49	Positive active power (decimal part)	Short	R/W	See the postscript to Table (8)
50~51	Active power (integer part)	Long	R/W	See the postscript to Table (8)
52	Active power (decimal part)	Short	R/W	See the postscript to Table (8)
53~54	Sensitive reactive power (integer part)	Long	R/W	See the postscript to Table (8)
55	Sensitively reactive power (decimal part)	Short	R/W	See the postscript to Table (8)
56~57	Capacitive reactive power (integer part)	Long	R/W	See the postscript to Table (8)
58	Capacitive reactive power (decimal part)	Short	R/W	See the postscript to Table (8)
59~60	Active power	Long	R/W	Secondary electric energy, 3 decimal places
61~62	A-phase active power	Long	R/W	Secondary electric energy, 3 decimal places
63~64	B-phase active power	Long	R/W	Secondary electric energy, 3 decimal places
65~66	C-phase active power	Long	R/W	Secondary electric energy, 3 decimal places
67~68	Total reactive power	Long	R/W	Secondary electric energy, 3 decimal places
69~70	A-phase total reactive power	Long	R/W	Secondary electric energy, 3 decimal places
71~72	B-phase total reactive power	Long	R/W	Secondary electric energy, 3 decimal places
73~74	C-phase total reactive power	Long	R/W	Secondary electric energy, 3 decimal places
78	Temperature measurement	Short	R	Signed short integer, with 1 decimal place
212	Relay 1 control item	Short	R/W	Range: 0 to 255
213	Relay 1 lower limit alarm value	Short	R/W	Range: 0 to 9999
214	Relay 1 upper limit alarm value	Short	R/W	Range: 0 to 9999
215	Relay 1 alarm return value	Short	R/W	Range: 0 to 9999
216	Relay 1 alarm delay value	Short	R/W	Range: 0 to 9999
217	Relay 2 control item	Short	R/W	Range: 0 to 255
218	Relay 2 lower limit alarm value	Short	R/W	Range: 0 to 9999
219	Relay 2 upper limit alarm value	Short	R/W	Range: 0 to 9999
220	Relay 2 alarm return value	Short	R/W	Range: 0 to 9999
221	Relay 2 alarm delay value	Short	R/W	Range: 0 to 9999
222	Relay 3 control item	Short	R/W	Range: 0 to 255
223	Relay 3 lower limit alarm value	Short	R/W	Range: 0 to 9999
224	Relay 3 upper limit alarm value	Short	R/W	Range: 0 to 9999
225	Relay 3 alarm return value	Short	R/W	Range: 0 to 9999
226	Relay 3 alarm delay value	Short	R/W	Range: 0 to 9999
227	Relay 4 control item	Short	R/W	Range: 0 to 255
228	Relay 4 lower limit alarm value	Short	R/W	Range: 0 to 9999
229	Relay 4 upper limit alarm value	Short	R/W	Range: 0 to 9999
230	Relay 4 alarm return value	Short	R/W	Range: 0 to 9999
231	Relay 4 alarm delay value	Short	R/W	Range: 0 to 9999
236	Transmitter Output 1 Control Item	Short	R/W	Range: 0 to 255
237	Transmitter Output 1 Upper Limit	Short	R/W	Range: 0 to 9999
238	Transmitter Output 2 Control Item	Short	R/W	Range: 0 to 255
239	Transmitter Output 2 Upper Limit	Short	R/W	Range: 0 to 9999
240	Transmitter Output 3 Control Item	Short	R/W	Range: 0 to 255
241	Transmitter Output 3 Upper Limit	Short	R/W	Range: 0 to 9999
242	Transmitter Output 4 Control Item	Short	R/W	Range: 0 to 255
243	Transmitter Output 4 Upper Limit	Short	R/W	Range: 0 to 9999

252	Transmitter Output 1 Output Value	Short	R/W	Range: 0~9999 (corresponding to 0-20mA or 0-5V). When the transmitter output is set to 0, the value can be written to control the corresponding transmitter output, and the current output value can be read.
253	Transmitter Output 2 Output Value	Short	R/W	
254	Transmitter Output 3 Output Value	Short	R/W	
255	Transmitter Output 4 Output Value	Short	R/W	

explain :

(1) The status byte of the switch input port (BIT0, BIT1, BIT2, BIT3) indicates the state of switch input ports 1, 2, 3, and 4 respectively, where 0 represents open state and 1 represents closed state. Writing this byte has no effect on the input ports. For the relay output status byte: During read operations, bits BIT0, BIT1, BIT2, and BIT3 indicate the output states of relays 1, 2, 3, and 4 (0 for open state, 1 for closed state). During write operations, bits BIT4, BIT5, BIT6, and BIT7 represent the write enable states for relays 1, 2, 3, and 4 (1 for enabling write control of corresponding relays, 0 for disabling). Bits BIT0, BIT1, BIT2, and BIT3 indicate the values of control relays 1, 2, 3, and 4 (1 for enabling control relays, 0 for disabling). Remote control is only valid when the current relay is in remote operation mode and the corresponding write enable bit is set to 1.

(2) The measured voltage is the secondary-side voltage value, fixed to one decimal place. The secondary-side voltage value equals the reading value divided by 10, while the primary-side voltage value equals the reading value multiplied by the PT ratio divided by 10.

(3) The measured current value is displayed on the secondary side with three decimal places. The secondary side current value equals the reading divided by 1000, while the primary side current value equals the reading multiplied by the CT ratio divided by 1000.

(4) The power and power factor symbol bit register contains seven bits (BIT0-7) in the low byte: BIT0-7 respectively indicate the sign of active power for phase A, phase B, phase C, total active power, and reactive power for each phase. A value of 0 denotes positive, while 1 denotes negative. The high byte's bits (BIT0-3) indicate the phase power factor sign (0 for inductive, 1 for capacitive) and the total power factor sign (0 for inductive, 1 for capacitive).

(5) The read power value is the secondary-side power value, fixed to one decimal place. The secondary-side power value equals the read value divided by 10, while the primary-side power value equals the read value multiplied by the PT ratio and CT ratio, then divided by 10.

(6) The power factor is fixed to 3 decimal places, and the power factor value is calculated as the reading divided by 1000.

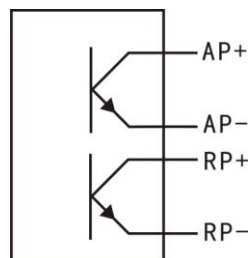
(7) Frequency is fixed to 2 decimal places, with the frequency value calculated as the read value divided by 100.

(8) The electrical energy value consists of three registers (Word0, Word1, Word2). The first two registers form a long integer representing the integer part, while the last register forms an integer representing the decimal part, which is a 3-bit decimal. The electrical energy value is calculated as $\text{Word0} \times 65536 + \text{Word1} + \text{Word2} / 1000$, with units of KWh or KVarh.

6. 2 Electricity Metering and Electric Pulse Output

The multifunctional power meter provides bidirectional active and reactive power measurement, with dual power pulse output channels and an RS485 digital interface for data display and remote transmission. The collector-level open-circuit optocoupler enables remote transmission of both active and reactive power, allowing remote computer terminals, PLCs, or DI switch acquisition modules to collect pulse counts for cumulative power measurement. The output method complies with national metrology standards (State Metrology Regulations: Pulse Error Comparison Method for Standard Meters).

(1) Electrical characteristics: The circuit diagram of the pulse acquisition interface is shown below, featuring optical coupling isolation with a collector and open-circuit output.



(2) Pulse constant: 3600 imp/kWh, meaning 3600 pulses are generated when the meter accumulates 1kWh. Note that 1kWh is the secondary measurement of electrical energy. For PT and CT, 3600 pulses correspond to $1 \text{ kWh} \times \text{voltage ratio (PT)} \times \text{current ratio (CT)}$.

(3) Application example: The PLC terminal uses a pulse counting device. Assuming that the number of pulses collected during a period of time t is N , and the instrument input is 10kV/100V, 400A/5A, then the accumulated electrical energy of the instrument during this period is $N/3600 \times 100 \times 80 \text{ kWh}$.

6. 3 Relay Output

The relay has a capacity of AC 250V/5A and DC 30V/5A.

The output module of relay has three working modes: electric quantity alarm mode, switch quantity control mode and remote control mode.

Remote Control Function: The 01H command reads the relay's output status, while 05H and 0FH commands control the relay's output status. The 10H command allows writing control information to the DO information register, enabling switching of digital input ports: writing 1 activates the corresponding port, and writing 0 deactivates it. For example, writing the binary code 00110001 activates port 1, deactivates port 2, and leaves relays 3 and 4 unaffected. To activate the remote control function, set the alarm parameter to 0.

In remote control mode, the relay operates in two modes: self-holding mode and pulse mode. When the delay time is set to 0, it operates in self-holding mode. When the delay time is set to a non-zero value, it operates in pulse mode, with the delay time equaling the pulse relay's action duration. If the relay exceeds the preset delay time, it automatically resets.

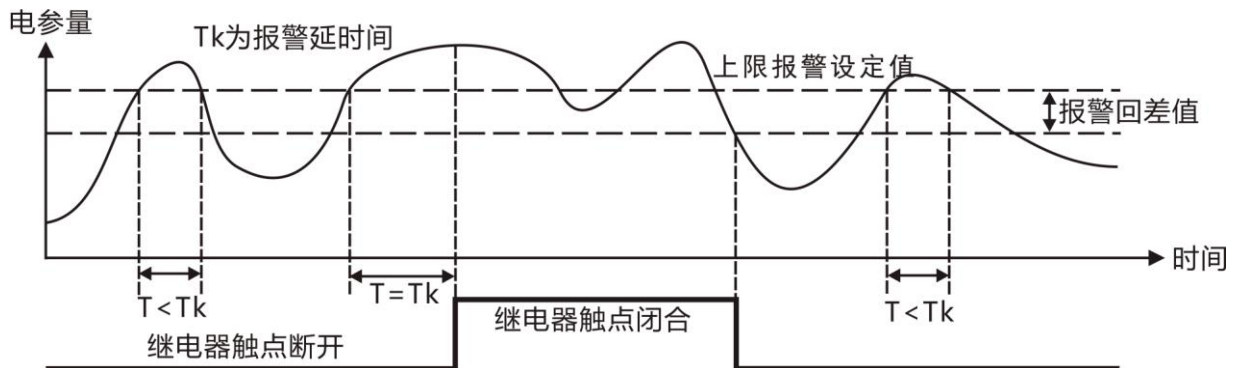
The alarm delay value means that the corresponding relay will act only after the alarm condition is satisfied for a set alarm delay time when the alarm mode is electric quantity alarm or switch quantity control.

Alarm Parameters: Define the electrical parameters to trigger alarms, including alarm modes (upper limit, lower limit, or both), lower limit threshold, upper limit threshold, and alarm return value. When the measured electrical parameter exceeds the alarm range, the corresponding switch output port will be activated. When the measured electrical parameter returns to normal range, the switch output port will be deactivated. Alarm parameters are configured based on secondary-side values. During setup, primary-side values must be converted to secondary-side values. For voltage alarms: Alarm value = Alarm value / PT ratio. For current alarms: Alarm value = Alarm value / CT ratio. For power alarms: Alarm value = Alarm value / (PT ratio × CT ratio). Programming example: For instruments with 10KV/100V and 400A/5A ratings, configure DO1 to trigger alarms when U_a exceeds 11KV and DO2 to trigger alarms when I_a exceeds 400A. The alarm settings are:

class	Warning condition	Relay control top	Alarm value
Relay 1	$U_a > 11KV$	65	110.0
Relay 2	$I_a > 400A$	71	5.000

Switching alarm: The output port is controlled by the input port, which can be set to either the output port being controlled by the input port's conduction or the input port being controlled by the input port's disconnection.

Alarm action diagram:



Switch output items (transmitter output items) reference table:

The alarm setting is 0, and the relay is in remote control mode.

Project	Digital output			Transmitter output			
	Lower limit alarm (switching value) Enter the report to notify the police	Upper limit alarm (switching value) Input Dis-connection Alarm	Upper AND LOWER LIMITS	0~20mA	4~20mA	0~10~20 mA	4~12~20 mA
U_a (Phase A voltage)	1	65	129	1	65		
U_b (B-phase voltage)	2	66	130	2	66		
U_c (C-phase voltage)	3	67	131	3	67		
U_{ab} (AB line voltage)	4	68	132	4	68		
U_{bc} (BC line voltage)	5	69	133	5	69		

Uca (CA line voltage)	6	70	134	6	70		
Ia (Phase A current)	7	71	135	7	71		
Ib (B-phase current)	8	72	136	8	72		
Ic (C-phase current)	9	73	137	9	73		
Pa (active power of A phase)	10	74	138	10	74	138	202
Pb (active power of phase B)	11	75	139	11	75	139	203
PC (C-phase active power)	12	76	140	12	76	140	204
P (Total Active Power)	13	77	141	13	77	141	205
Qa (A-phase reactive power)	14	78	142	14	78	142	206
Qb (B-phase reactive power)	15	79	143	15	79	143	207
Qc (C-phase reactive power)	16	80	144	16	80	144	208
Q (Total Reactive Power)	17	81	145	17	81	145	209
Sa (A phase in power)	18	82	146	18	82		
Sb (B phase is the power)	19	83	147	19	83		
Sc (C phase power)	20	84	148	20	84		
S (Total Apparent Power)	21	85	149	21	85		
PFa (Phase A power factor)	22	86	150	22	86	150	214
PFb (Phase B Power Factor)	23	87	151	23	87	151	215
PFc (Phase Factor of Current)	24	88	152	24	88	152	216
PF (Power Factor)	25	89	153	25	89	153	217
Freq (frequency)	26	90	154	26	90	154	218
Simultaneous monitoring of Ua, Ub, and Uc	45	109	173				
Simultaneous monitoring of Uab, Ubc, and Uca	46	110	174				
Simultaneous monitoring of Ia, Ib, and Ic	47	111	175				
Active power of reverse phase	48	112	176				
DI1 (Switch Input 1)	49	113					
DI2 (Switch Input 2)	50	114					
DI3 (Switch Input 3)	51	115					
DI4 (Switch Input 4)	52	116					

6. 4 On-off Input

The switch input module employs a dry contact switch signal input method. The instrument is equipped with an internal +15V working power supply, eliminating the need for external power. It can monitor parameters such as fault alarm nodes, switch status (open/closed), trolley position, and capacitor engagement status in capacitor compensation cabinets. Status information can be remotely transmitted to intelligent monitoring systems via communication interfaces. When paired with remote control/alarm relay functions, it enables automatic switch operations. The monitored switch status can be configured as relay output conditions to implement interlocking and mutual locking functions. The RS485 interface allows reading switch input interface status through the 02H command.

6. 5 Analog Transmitter Output

The multifunctional power meter features 4 analog signal output channels, each configurable to select any of 26 electrical parameters as the output signal. The meter's built-in transmitter function supports analog signal outputs (0~20 mA, 4~20mA, 0~10~20mA, or 4~12~20mA), with customizable parameter mappings. The analog output can also be remotely controlled, where the transmitter's low output corresponds to remote control setting 0 and the high output to setting 9999.

Electrical parameters: Output ranges 0~20mA, 4~20mA, 0~5V, and 0~10V with Class 0.5 precision; load resistance <500Ω.

The transmitter output item in the menu allows you to select one of 26 electrical parameters as the output, with options ranging from 0 to 20mA.

The transmitter output ranges are 4~20mA, 0~10~20mA, or 4~12~20mA. Refer to the transmitter output item comparison table for the relationship between the transmitter output items and electrical parameters. The transmitter output range is used to set the measurement value of the electrical parameter corresponding to the transmitter output upper limit of 20mA. The measurement value here is the secondary side value. The method for converting primary side values to secondary side values is the same as the setting of relay alarm output values. For details, refer to the alarm output value settings. The power factor transmitter output is fixed at 0.500~1.00 0, corresponding to DC0~20mA or 4~20mA.

The bidirectional transmitter outputs range from 0 to 10 mA or 4 to 12 mA. For frequency bidirectional transmitter outputs, the output is centered at 50 Hz. If set to 5.00 Hz, the corresponding output range is 45.00Hz~50.00Hz~55.00 Hz. For active (or reactive) power bidirectional transmitter outputs, a set value of 3300W produces an output range of -3300W to +3300 W. For power factor bidirectional transmitter outputs, the fixed range is C0.500 to 1.000 and L 0.500.

6. 6 Temperature Measurement

The temperature measurement is provided by DS18B20 sensor, with a measurement range of -40~120°C and an accuracy of $\pm 0.5^\circ\text{C}$.

VII. USER TYPE SELECTION

7. 1 Naming Convention

ZH194F-①S②-DO③-DI④-AO⑤-⑥

- ① Instrument panel size code: 120X120 panel size 2; 96X96 panel size 9
- ② Display type code: LED digital tube displays as 4; LCD liquid crystal displays as Y.
- ③ Switch output indicator: Displays the number of channels with switch outputs. This field is blank if no output is present. For example, 'with 2 switch outputs' means DO2.
- ④ Switch input indicator: Displays the number of channels with switch inputs. This field is blank if no input is present. For example, 'with 4 switch inputs' is represented as DI4.
- ⑤ Analog output indication: Displays the number of channels with analog output. This field is blank if no output is available. For example, '3 channels with analog output' is AO3.
- ⑥ Temperature measurement indicator: Indicates whether the device has temperature measurement capability. T indicates enabled, while empty indicates disabled.

Example: A multifunctional power meter with an LCD display, featuring a 96×96 panel and two analog outputs. Model: ZH194F-9SY-H-DO2-DI2.

7. 2 Function Declaration

For special requirements, please contact directly for customized specifications.

Email: 275947102@qq.com



LCD meter appearance



LED meter appearance